

AI-DRIVEN MULTI-CLINIC HEALTHCARE APPOINTMENT AND RESOURCE OPTIMIZATION PLATFORM

PHASE 1 REPORT

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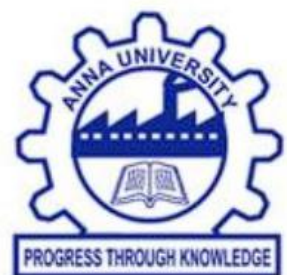
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In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

ANNA UNIVERSITY, CHENNAI

OCTOBER – 2026

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Report titled "**AI-Driven Multi-Clinic Healthcare Appointment and Resource Optimization Platform**" is the bonafide work of **Sheryl Katrina M (230701310), Shalini R K (230701304), Senthamizh Selvi V (230701301), Sangamithran A (230701283)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

This project addresses the following Sustainable Development Goal:

Goal- No :3 Good Health and Well-being.

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ABSTRACT

Healthcare clinics are essential for offering affordable and effective medical services. However, they do encounter challenges like long waiting times, ineffective appointment scheduling, uneven workload of doctors, manual queue management, and fragmented patient and pharmacy records. - Reduced patient satisfaction, increased administrative burden and impact on the efficiency of the overall health delivery. The proposed project, Med-link proposes an AI-powered clinic management framework which incorporates intelligent patient triage, adaptive appointment scheduling, digital consultation management and pharmacy inventory management. The system uses AI-based symptom analysis to assist in deciding patient priority and sort the queue according to urgency of condition. An adaptive scheduling module optimizes appointment allocation and balances doctors' workload considering patient priority, doctors' availability and expected consultation duration. The platform is web-based and provides patients with information about appointments, queue status, and automated notifications. The digital patient's records and consultation details can be easily managed by healthcare professionals. The pharmacy module monitors medicine availability and helps to identify possible stock shortages. Unlike conventional clinic management systems that rely on manual scheduling and record keeping, MedLink uses intelligent decision-making, automated workflow management and centralized healthcare data management. The framework helps to reduce the waiting time of patients, improve the use of resources, improve the distribution of work and improve the operational efficiency of the clinic. MedLink's combination of artificial intelligence and digital healthcare management offers a new, scalable approach to the real-world problems of today's clinical settings.

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ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J. Manoranjini.**, Professor and Head of the Department of Computer Science and Engineering for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, **Mr. N. Karthikeyan., M.E.**, Assistant Professor, Department of Computer Science and Engineering for his valuable guidance throughout the course of the project. We thank our Project Coordinator, **Dr.S.MADHUSUDHANAN,Ph.D.**, Associate Professor, Department of Computer Science and Engineering for his useful tips during our review to build our project.

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LIST OF ABBREVIATION

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
Bi-LSTM	Bidirectional Long Short-Term Memory
DPI	Dots Per Inch (Image Resolution)
EHR	Electronic Health Record
ETA	Estimated Time of Arrival
FCFS	First-Come, First-Served (Baseline Model)
FEFO	First-Expired, First-Out
HTML	HyperText Markup Language
MAE	Mean Absolute Error
NLP	Natural Language Processing
OPD	Outpatient Department
PNG	Portable Network Graphics
Q-Learning	Model-Free Reinforcement Learning Algorithm
REST	Representational State Transfer
RMSE	Root Mean Square Error
UI / UX	User Interface / User Experience
CDN	Content Delivery Network

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Clinics for healthcare play a crucial role in providing timely and accessible medical care to patients. Even so, many clinics still rely on traditional and manual means to handle pharmacy operations, consultations, patient queues and appointments. With the increasing number of patients these methods can result in inefficient scheduling, long waiting time, uneven distribution of patients to doctors and difficulties in keeping patient records accurately. Additionally, without intelligent prioritization, healthcare staff may also have difficulty identifying and treating patients according to the urgency of their conditions.

The proposed AI-Powered Clinic Management System is an attempt to solve these issues by combining Artificial Intelligence with essential clinic management operations. The platform offers intelligent patient triage, adaptive appointment scheduling, automated queue management, digital consultation management and pharmacy inventory management all in one platform. The AI based triage module helps in prioritizing the patients based on their symptoms and other relevant information. The scheduling module schedules the appointments effectively taking into account multiple variables, including priority of the patient, availability of the doctor, expected time for consultation, etc. Plus, automated notifications and reminders can reduce no-shows and increase communication between patients and the clinic.

The system will integrate intelligent decision making with centralized digital management, to improve the overall efficiency and reliability of clinic operations. Healthcare providers can manage patient information, consultations, appointments and medicine availability from one platform and patients can have a better organized appointment and queue experience. The system aims at reducing administrative workload, shortening unnecessary waiting times, improving the distribution of the doctors' workload and supporting the effective use of the clinic's resources. Therefore, the proposed solution offers a scalable and intelligent way to modernize traditional clinic management and improve the quality of healthcare service delivery.

1.2 OBJECTIVE

The main aim of the AI-Powered Clinic Management System is to design and implement an intelligent healthcare management system that uses Artificial Intelligence to solve the problems that currently affect the operation of traditional clinics.

1. Intelligent Patient Triage: To develop an AI-driven triage module that evaluates patient symptoms and pertinent details to determine patient priority and help healthcare workers manage urgent scenarios effectively.

2. Adaptive Appointment Scheduling: To develop an intelligent scheduling mechanism to optimize the appointment allocation and improve the distribution of doctor workload by taking into account patient priority, doctor availability and expected consultation time.

3. Automatic Queue Management: Develop a digital queue management system which prioritizes and organizes patient flow based on appointment status and priority, reducing unnecessary wait times and improving the patient experience.

4. Digital Management of Consultations: To create a central digital platform that enables healthcare professionals to efficiently access and manage patient records, medical histories, appointment details and consultation information.

5. Pharmaceutical Inventory Management: This will help design an inventory management module to facilitate pharmacy work and timely replenishment by monitoring the availability of the medicine, tracking the stock levels and identifying the potential shortage.

6. Automated Notifications and Reminders:

An automated appointment confirmation and reminder system can be implemented to improve patient communication and decrease missed appointments.

The achievement of these goals will convert the traditional clinic management into a more patient-centered, efficient and intelligent health care service which will enhance the utilization of resources,

reduce the administrative load, reduce the waiting time and improve the general quality of the clinic operations.

1.3 EXISTING SYSTEM

At present, the clinic management system is mainly based on manual and simple electronic equipment to perform the activities of patient registration, appointment scheduling, queue management, consultation records and pharmacy. These systems can support the routine administrative tasks but they do not have the ability to make intelligent decisions to prioritize patients effectively based on their symptoms and urgency. Appointment schedules are often developed based on fixed time slots and doctor availability that can lead to overlapping appointments, skewed doctor workloads and increased patient waiting times especially during peak hours. Also, the dynamic management of patient queues is not well supported by existing systems as clinical priorities change.

The vast majority of clinics still rely on receptionists and healthcare staff to manually manage and monitor patients as a core part of their operations. These techniques are labor-intensive, subjective and difficult to scale as patient volume increases. Finally, the employment of distinct systems or physical documents for the storage of patient records and consultation data can pose challenges in information retrieval and coordination efforts. Stock in pharmacies is usually tracked manually leading to poor inventory management and slow detection of medicines that are low in stock. Some clinics have moved to electronic health records and online appointment systems, but they are mostly digitizing existing processes, not enabling intelligent automation.

Another significant limitation of current systems is the absence of AI-assisted flexibility and embedded decision support. Most traditional platforms do not use patient, appointment or operational data to optimize resource allocation and scheduling. They also don't have much automation for patient communication and appointment reminders which can cause missed appointments. The result is less patient satisfaction, underused clinic and doctors, increased waiting time and administrative workload. These limitations reveal the need for intelligent and integrated system of clinic management which is feasible to real-time healthcare requirements.

1.4 PROPOSED SYSTEM

It is developed to overcome the drawbacks of traditional clinic management systems and provides an intelligent and integrated healthcare framework. The proposed system combines AI-based Patient Triage, Adaptive Appointment Scheduling, Automated Queue Management, Digital Consultation Management and Pharmacy Inventory Management in a single platform. The Patient Triage Module assesses the patient's symptoms and relevant information to rank the cases in order of urgency, helping healthcare professionals to handle patients efficiently and provide urgent care in a timely manner.

The Adaptive Appointment Scheduling Module generates an optimal appointment schedule based on the priority of the patient, the availability of the physician and the expected time of the consultation. Plus there's an automatic queue regulation system to make sure patient flow is optimal and unnecessary waiting times are reduced. Appointment reminders and notifications enhance patient communication and help to reduce no-shows. These features make it easier for doctors to cope with their workload when there are many patients.

The proposed system is a single digital platform to maintain the patient records and medical history, appointments and consultation details. Healthcare providers have easy access to and can easily update patient information, increasing the availability of information and reducing the dependence on manual records. Pharmacy Inventory Module: The module helps to manage inventory effectively and to replenish stock on time, by monitoring the medicine stock levels and identifying probable shortages.

The suggested system is a common digital platform to keep patient records, medical history, appointment and consultation details. This allows healthcare providers to easily access and update patient information. This leads to better availability of information and less dependence on manual records. The Pharmacy Inventory Module is used to manage the inventory in an efficient way and to ensure that the stock of medicines is replenished in time by monitoring the levels of stock of medicines and identifying possible shortages.

CHAPTER 2

LITERATURE SURVEY

The rapid development of digital healthcare technologies, machine learning and Artificial Intelligence has led to the creation of intelligent healthcare management systems intended to improve operational efficiency and patient care. Existing clinic management solutions generally focus on appointment booking, patient registration, record maintenance and are not well addressed in terms of adaptive scheduling, intelligent patient prioritization, patient flow and utilization of resources. In this context, research papers on AI-based triage, scheduling of healthcare appointments, prediction of patient no-shows and decision support in clinical care are of high relevance. This chapter offers a review of the key works in these areas, with a focus on their methodologies, advantages and disadvantages and their role in the development of the proposed AI-Powered Clinic Management System.

2.1. LITERATURE REVIEW

A Systematic Review, “Artificial Intelligence for Better Triage Decisions in Healthcare,” summarizes the use of machine learning and artificial intelligence (AI) technologies to improve patient triage. Artificial intelligence can help health professionals to make triage decisions more consistent, efficient and accurate, a review of studies between 2020 and 2025 has found. The reviewed techniques were hybrid techniques, deep learning, machine learning and natural language processing. The study also highlighted challenges in terms of ethical issues, usability, trust and adaptation to different healthcare settings. The proposed system demonstrates the potential of AI-based triage to support healthcare staff in the prioritization of patients and the need of explainable and responsible decision support. In the paper “Appointment Scheduling Problem in Complexity Systems of the Healthcare Services: A Comprehensive Review” a comprehensive review of the approaches to scheduling of healthcare appointments is provided. The research looked at over 150 papers, examining variables such as patient flow, patient preferences, random arrivals and service times. In this paper we studied the reduction of patient waiting time with different methods such as mathematical optimization, simulation models, markov models and artificial intelligence. optimizing the use of health care resources. According to studies, smart scheduling can boost patient satisfaction, reduce idle time and eliminate operational inefficiencies.

Artificial Intelligence's Role in Improving Triage Decisions in Healthcare Settings: A Systematic Review of Machine Learning and AI Methods to Improve Patient Triage A review of studies from 2020-2025 indicates that AI-powered systems can assist health professionals to make more consistent, efficient and accurate triage decisions. The techniques reviewed included deep learning, machine learning, hybrid methods, and natural language processing. However, the study also pointed out challenges related to ethical concerns, usability, trust and adaptation to different health care settings. The proposed system demonstrates the potential of AI-based triage to support healthcare staff for patient prioritization, as well as the importance of explainable and responsible decision support.

The paper “Appointment Scheduling Problem in Complexity Systems of the Healthcare Services: A Comprehensive Review” offers a broad review of the approaches for scheduling health-care appointments. The study reviewed over 150 research papers and considered a number of variables including patient flow, patient preferences, random arrivals and service times. Different methods have been reviewed on the reduction of patient wait time, including mathematical optimization, simulation models, Markov models, and artificial intelligence . The study “Artificial Intelligence Applications Using Patient-Generated Health Data for Pre-Care Processes in Elective Healthcare: A Systematic Review” examines the use of AI on patient-generated health data to support pre-care processes such as triage, symptom assessment and medical history collection. The study demonstrates how standard artificial intelligence can facilitate the preparation of patient information ahead of the clinical encounter. This approach is relevant to the proposed system as the collection and analysis of patient symptoms prior to consultation can aid with more organized patient prioritization and improve the efficiency of clinic workflows.

2.2 COMPARISON AND DISCUSSION

Literature review showed that machine learning and artificial intelligence are very useful in the improvement of healthcare management, especially in patient triage, scheduling appointments and prediction of missed appointments. In contrast, intelligent systems can evaluate historical and real-time data to help make better decisions, whereas current appointment systems are mostly based on strict schedules and human intervention. Studies have indicated machine learning and deep learning techniques can facilitate patient prioritization and improve consistency in healthcare decisions for AI based triaging. However, the reviewed literature also emphasize the importance of trust, interpretability,

ethical issues and supervision by humans when applying AI in healthcare. Hence the proposed system is aimed to use AI as a decision supporting mechanism rather than replacing the health care professionals. The literature on appointment management shows the effectiveness of predictive models and optimization techniques to reduce the waiting time, optimize the use of resources and manage the patient no-shows. But, many approaches only address one problem, e.g., scheduling or no show prediction, but not combining more than one clinic operation into one platform. This gap is filled with the proposed AI-Powered Clinic Management System that combines patient triage, flexible appointment scheduling, automated queue management, digital consultation management and pharmacy inventory management into a single unified framework. This combination makes it possible to feed information generated by one module back into the decision making of other modules. For example, patient priority can be used to arrange appointments and appointment information can be used for queue management and automated reminders; Overall, the comparison shows that the current research offers strong individual solutions, but there is scope for integrating them into a smart, scalable and complete clinic management framework. To fill this gap, in this paper, we propose an integrated system of clinic management functions with AI based decision support .

2.3. CONCLUSION

To summarize, the reviewed studies indicate the increasing role of Artificial Intelligence and machine learning in modern healthcare management. AI-based triage can help in prioritizing patients, predictive scheduling can help improve the use of resources, and machine learning-based no-show prediction can help reduce missed appointments. However, existing approaches are usually limited to specific healthcare processes and do not offer a comprehensive solution for the overall clinic operations.

The proposed AI-Powered Clinic Management System bridges this gap by providing a single platform for intelligent patient triage, dynamic appointment scheduling, automated queue management, digital consultation management and pharmacy inventory management. The system intends to integrate these functionalities in an effort to develop a more efficient, flexible, patient-centered and data-driven approach to the management of the clinic, helping health care professionals to take better decisions at the operational level.

CHAPTER 3

SYSTEM DESIGN

The system design process translates the requirements of AI-Powered Clinic Management System into a framework. It defines the system architecture, modules and data flow. The system consists of AI-based Patient Triage, Adaptive Appointment Scheduling, and Automated Queue Management. It also has module for Digital Consultation, Pharmacy Inventory & Automatic Notification. This design ensures efficient data management, intelligent decision making, reliability, scalability and efficiency of the clinic improved.

3.1 SYSTEM ARCHITECTURE

The AI-Powered Clinic Management System is a modular and integrated healthcare framework connecting patients, doctors and clinic assistants through dedicated user applications and a centralised backend. The frontend layer consists of Patient Mobile App, Doctor Mobile App and Clinic Assistant Portal with role specific functionalities. Patients are able to register, authenticate, make appointments, track queues, view health records, and get notifications. Doctors can handle appointments, availability, consultations, clinical notes and patient history, while clinic assistants can handle clinic operations, appointments, patient check-in, queues, inventory and reports. All applications communicate to backend via secure HTTPS REST APIs and Socket.[IO](#) for real-time updates.

The Backend and Core Services comprise the main processing tier of the system and are responsible for managing the core clinic operations. This layer services include Authentication and Authorization, User Management, Appointment Service, OPD Queue Engine, Recommendation Engine, Notification Service, File Storage Service, Audit Logging Appointment Service for booking, rescheduling and cancelation OPD Queue Engine: Real-time queue updates and patient flow management. Recommendation Engine provides recommendations of departments and procedures Notification Service manages notifications by app, email, SMS These services are connected through the MediLink Backend API, which contains the RESTful API layer, business logic layer, service integration layer, [Socket.IO](#) event handler, and data access layer.

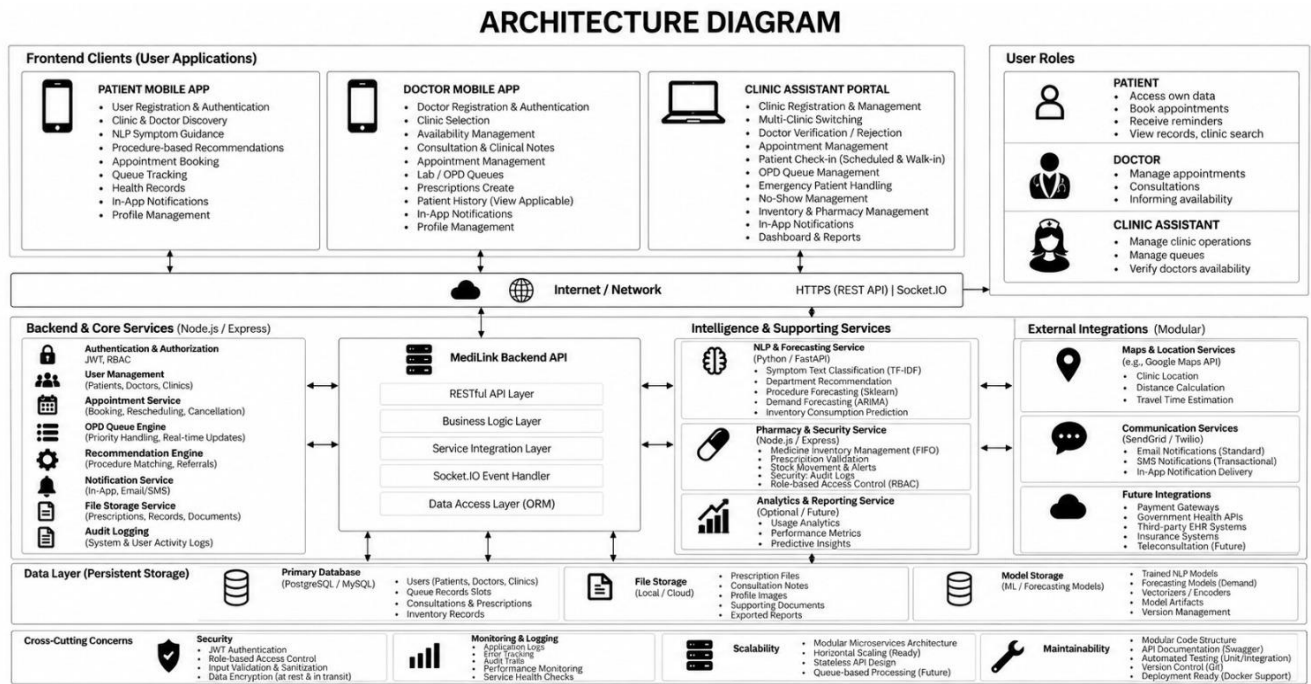


Figure 3.1: End-to-End System Architecture of the MedLink Healthcare Optimization Platform

The Intelligence and Supporting Services offers AI-enhanced capabilities to enhance decision-making and operations at the clinic level. NLP and Forecasting Service is a python FastAPI service that performs symptom classification, department recommendation, procedure forecasting, demand forecasting and inventory consumption prediction. The system also provides a Pharmacy and Security Service for medicine inventory management, prescription validation, stock movement and alerts, security audit logs and role based access control. An Analytics and Reporting Service complements the usage analytics, metrics for performance and predictive insights. The system is also integrated with external services such as Maps and Location Services for clinic location, distance calculation and travel-time estimation; and communication services such as e-mail, SMS and in-app notification. The Data Layer is where the information the system generates and processes is kept. The main database contains users, doctors, patients, clinics, queue records, consultations, prescriptions and inventory records. File storage stores prescription files, consultation notes, profile images, supporting documents, and exported reports Model storage stores trained NLP and forecasting models, vectorizers, model artifacts, and version information Security, monitoring, logging, scalability and maintainability are examples of cross-cutting concerns that are applied across the architecture. JWT authentication . Role based access . Input validation . Data encryption . Application logs . Performance .

3.2 SYSTEM REQUIREMENTS

The AI-Powered Clinic Management System includes intelligent patient triage, adaptive appointment scheduling, queue management, digital consultation, pharmacy inventory management and automated notifications. The system requirements are divided into functional and non-functional requirements for the proper development, deployment and operation of system. Hardware and Software Requirements To ensure smooth development, deployment, and operation of the AI-Powered Clinic Management System, the following hardware and software specifications are recommended.

3.2.1 Hardware Requirements:

1. Server / engine side processing

- a. AI Process Oriented Back End Services b. Multi core processor (Intel i5/i7 or equivalent)
You need at least 16 GB RAM to run machine learning algorithms smoothly.
- b. Minimum 16 GB RAM for smooth working of AI Models, Databases and multiple services.
- c. Storage of patient records, consultation data, pharmacy data and AI models SSD (min. 500 GB).

1. User Devices

- a. **Smartphones or tablets** for patients and doctors to access their respective mobile applications.
- b. **Desktop/laptop computers** for clinic assistants to manage appointments, queues, patient records, inventory, and reports.

2. Networking

- a. Stable **internet connectivity** for communication between frontend applications, backend services, and external services.
- b. Secure network connectivity to support **HTTPS REST APIs and Socket.IO** for real-time communication and queue updates

3. Supporting Devices

- a. A reliable display and input system for clinic staff to monitor **appointments, patient queues, consultations, and pharmacy inventory.**

- b. Adequate local or cloud-based storage for maintaining **documents, prescriptions, consultation notes, and exported reports.**

The hardware configuration is made to handle the information, scheduling, live queues AI calculations and pharmacy activities for the AI-Powered Clinic Management System. It has the processing power , enough memory , plenty of storage and reliable network connectivity . This allows several users to work together without any problems at the same time. The system can also grow in the future by adding users, clinics, services or new AI-based features.

3.2.2 Software Requirements:

1. Operating System

- a. Windows 10/11 or Linux (Ubuntu 20.04 or later) for development, testing, and server environments.

2. Development Tools and Languages

- a. Design and development of AI based patient triage, prediction and processing modules in python 3.x Thus we need to develop systems that will be able to assess the status of the patient fast and accurately. The modules are designed to deal with data volumes and give instant decision making to help healthcare staff. There are tons of libraries that can help you out with machine learning and data analysis . Python is a good choice for this work.
- b. Development of front end application and interactive web interface in Javascript and Typescript. The instruments are intended to be easy for the patient and medical staff Access to health information easily. Responsive and interactive UIs. When using Typescript you get structure and safety in your code making it easier to maintain and scale.
- c. Node.js with Express.js for developing backend services & REST APIs. This allows the database and the front-end to talk to each other seamlessly. The backend is responsible for authentication, data processing and API calls. Express.js – A fast, unopinionated, minimalist web framework for node. React.js for Web apps and Clinics Management Portals 0 4.
- d. These applications are available on any device from desktop to tablet to smartphone. React.js gets you to Render. It gives a smooth interaction to the user. Staff coordination, appointment scheduling, and storage of patient records are possible with clinic management portals.

3. Libraries and Frameworks

- a. ML Classification of symptoms, Recommendations and Prediction Analysis using Scikit-learn & other python libraries.
- b. Data processing and cleaning
- c. Data preparation and analysis with Pandas, Numpy Communication Backend: Socket IO for real time queue notifications and updates.

4. Database Management System

- a. Patient Information, Doctor Information, Appointments, Consultations, Prescriptions and Inventory in PostgreSQL / MYSQL database.
- b. Cloud/File storage To upload documents like consultations, prescriptions, reports and any other supporting documents.

5. Additional Tools

- a. Git / Github Source control and collaboration changes
- b. Swagger API Reference Server-side Service OpenAPI for documentation & testing RESTful
- c. External API's: Maps, communications Location services API's Email, SMS & Alerts Distance Calculation.

The proposed software stack will help to ensure that the AI-Driven Clinic Management System can efficiently facilitate the intelligent healthcare operations, real-time communication and secure data management. The system applies modern development frameworks, machine learning libraries, database technologies and third-party services to manage patients, schedule appointments, monitor queues, record consultations and control pharmacy operations. Additionally, this architecture will allow for future integration of other AI models, services and features, it will be a scalable, maintainable and adaptable system for future healthcare needs.

3.3. SYSTEM DESIGN

The design model of the AI-Powered Clinic Management System shows the movement of patient and clinic data through a sequence of structured processing stages, from raw inputs to well-informed healthcare decisions. The system gathers patient symptoms, appointment details, doctor availability, consultation records and pharmacy data through the respective user applications. Once validated and pre-processed, this data is fed into the AI-based Patient Triage Module and the Adaptive Appointment Scheduling Module which work together to decide the priority of the patient and allocate the appointments in the most efficient way possible. The Automated Queue Management Module then processes their outputs and manages the patient flow. The Digital Consultation and Pharmacy modules manage clinical records and medicine inventory, respectively. The Notification and Analytics services analyse the data to generate real-time notifications, timely alerts and valuable operational insights. With this design, smooth flow of data and intelligent decision making, real time responsiveness and reliable clinic management.

3.3.1 Activity Diagram:

Figure 4 shows the activity diagram of the AI-Powered Clinic Management System. The process begins with patient registration which means collecting details like symptoms, appointment details, medical history etc. These details are validated and pre-processed. This processed information is sent to the AI based Patient Triage Module and Adaptive Appointment Scheduling Module to decide the priority of the patient and to assign suitable appointment slots. The Automated Queue Management Module then controls the flow of patients based on their priority and appointment status. The Digital Consultation Module is used for storing consultation records and prescriptions. The Pharmacy Inventory Module maintains the medicine stock and sends alerts whenever required and the Notification Module sends appointment reminders and real-time updates to users. This workflow facilitates easier patient management, shorter waiting time and better coordination during all clinic operations.

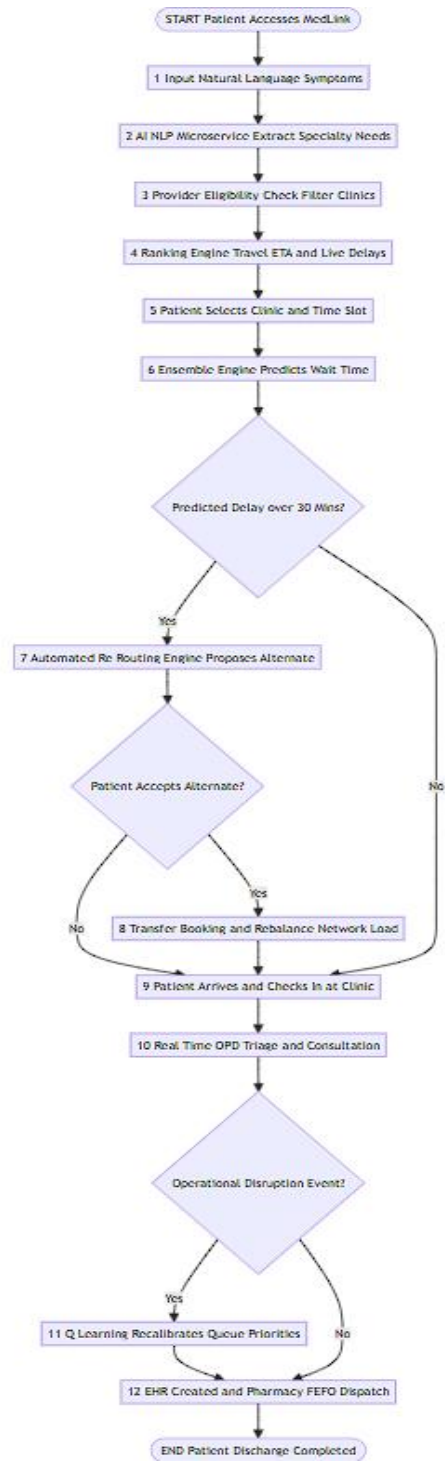


Figure 3.2: End-to-End System Workfl

3.3.2 Data Flow Diagram

The data flow diagram shows the conversion of patient and clinic data into useful healthcare outputs. The backend first validates and processes the data coming in from patients, doctors, and clinic assistants. The processed data is then analyzed by the Patient Triage and Appointment Scheduling modules and their results are fed into the Queue Management, Digital Consultation and Pharmacy Inventory modules. Finally, the information is stored and put to use for notifications, alerts, and analytics.

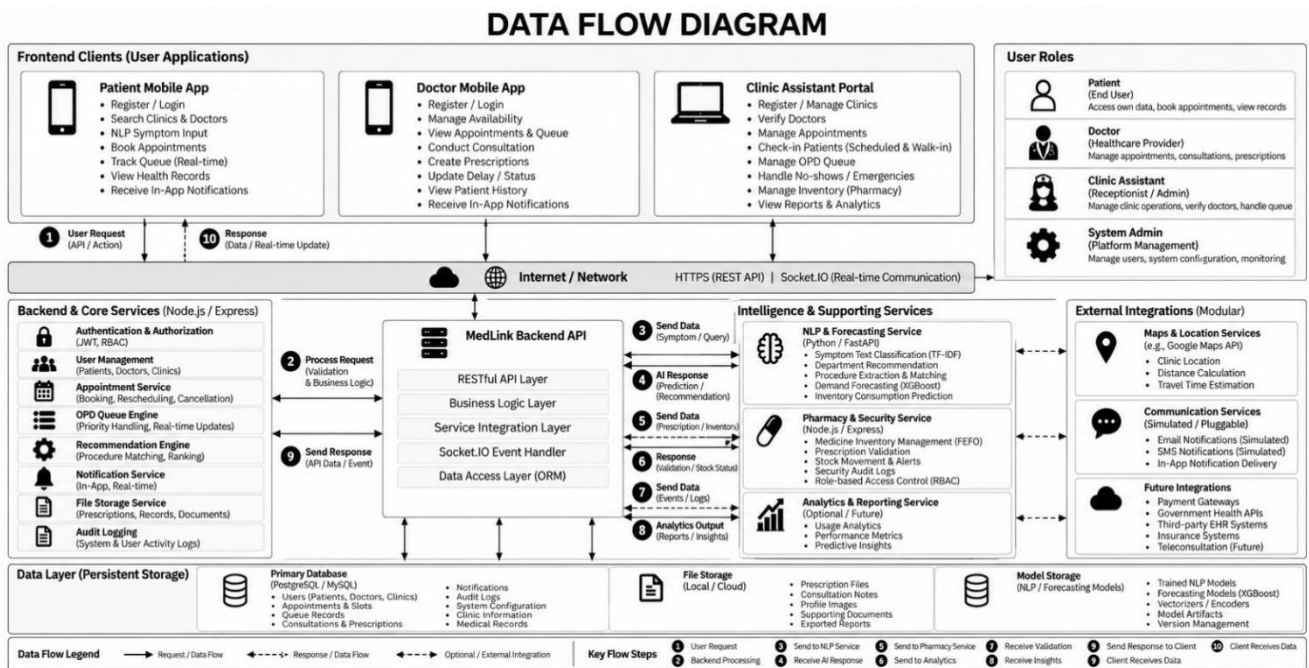


Figure 3.3: End-to-End Data Flow Diagram of the MedLink Platform.

The AI-Powered Clinic Management System is an end-to-end AI based solution with Patient Triage, Adaptive Appointment Scheduling and Predictive Analytics in order to improve clinic operations. The Patient Triage module analyzes patient symptoms and related information to assess the urgency of each case and to recommend the appropriate department. The Appointment Scheduling module allocates appropriate slots considering the priority of the patient, availability of doctor and estimated time of consultation. These smart modules work with the Queue Management system to help manage patient flow and reduce unnecessary waiting time.

Patient and clinic data is cleaned, validated and normalized before any processing. Feature extraction is applied where necessary. This gives the AI modules a steady and dependable input to work with. The system employs predictive techniques for functions such as procedure prediction, demand prediction, and inventory consumption prediction. The results are then integrated with the backend services for appointment management, queue organization, consultation handling and pharmacy operations. The appointment and queue information also produces automated notifications and reminders that improve communication with patients and helps to reduce missed appointments.

And the methodology is also built with scalability, security and modularity in mind, so that individual modules can be updated or improved without disturbing the rest of the system. APIs and event-based services for real-time communication deliver timely queue updates, notifications, and operational monitoring. Information can be easily extracted and analyzed when needed due to the centralized patient, appointment, consultation, prescription and inventory records. Combined, these methods make for an intelligent, reliable and flexible way of managing the modern clinic, helping to cut down on administrative workload, make better use of resources and improve the overall patient experience.

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

4.2 MODULES

MedLink Healthcare AI Platform is designed as a complete digital healthcare ecosystem that connects patients, doctors, clinic staff, pharmacy operations and administrators on the same platform. It combines web and mobile interfaces with a distributed backend, real-time communication services, artificial intelligence components, and persistent healthcare data management.

It has a modular architecture that separates the major functional responsibilities but allows the modules to communicate using REST APIs, WebSocket based real-time events, and shared database services. This approach makes the system easier to maintain, scale, and secure, and it allows each healthcare function to be developed independently.

The major modules of the MedLink system are:

1. User Authentication & Role Management Module:
2. Clinic Discovery& Clinic Recommending Module
3. Doctors availability & appointment module:
4. OPD Live Queue Management System
5. AI Module in Clinical Triage & Symptom Checker
6. Pharmacy stock and demand forecast module
7. E-Prescription & Patient Health Record Module
8. Notification and Communications Module
9. Administrative Dashboard and Analytics Module

4.2.1 USER AUTHENTICATION AND ROLE MANAGEMENT MODULE

The User Authentication and Role Management Module forms the security backbone of the MedLink platform. Because the system deals with healthcare information, access to application features and patient information must be controlled according to the authenticated user's role.

There can be different types of users in the system, such as patients, doctors, clinic staff, administrators. Authentication is performed using JSON Web Tokens (JWT) to enable the backend to verify every authenticated request before granting access to protected resources. The module handles user authentication, authorization, identity resolution and role-based access control. Passwords are securely hashed and each authenticated request includes the authorization information needed to identify the user making it.

Role-based authorization makes sure users can only perform the operations that belong to their responsibilities. For example, patients can manage their own appointments and health records, doctors can access the relevant consultation and patient information, clinic staff can manage queues and appointments, and administrators can reach the management and analytics functions.

The module also supports verification workflows and related communication, such as OTP-based verification.

Main features

- User sign up and authentication
- Session authorization based on JWT
- Role based access control
- Hashing passwords
- One-time password / verification flows
- Protected API endpoints
- Resolution of user identity
- Information for doctor verification
- Access to secure healthcare records

In addition, the implementation is more modular and the code is easier to maintain because the authentication middleware, authentication routes and authentication services are decoupled from the rest of the business logic. In short, the User Authentication and Role Management Module is the security backbone of the MedLink Platform. It utilizes JWT based authentication, secure password hashing and OTP verification to make sure that only genuine users can enter the system.

4.2.2 CLINIC DISCOVERY AND RECOMMENDATION MODULE

The Clinic Discovery and Recommendation Module assists patients in locating appropriate healthcare facilities according to their location, availability, waiting time and other relevant factors. The platform combines location and geography services with the data of the registered clinics. Its architecture provides geographic discovery with external location data sources and routing services, allowing clinics to be found by proximity and likely travel involvement.

The recommendation engine takes a variety of factors into account, not just physical distance. At present, the recommendation mechanism takes into account factors such as:

Distance Estimated travel time

- Doctor availability
- Average waiting time
- Rating of the clinic · Patient choice

The engine attaches a configurable weight to each of these factors and merges them into a single score that is used to make a recommendation.

The current weighting implemented in the project is:

Factor	Weight
Distance	0. 30
Estimated Travel Time	0. 20
Availability	0. 20
Waiting Time	0. 15
Rating	0. 10
Patient Preference	0. 05

The score obtained enables to present the clinics on several operational and geographical criteria. This module is of utmost importance as a clinic in close geographical proximity may not be the most practical option if there is no doctor availability or the expected waiting time is high.

Major functionalities

- Location-based clinic discovery
- Clinic filtering
- Distance calculation
- Estimated travel-time integration
- Doctor availability consideration
- Waiting-time consideration
- Multi-factor recommendation scoring
- Ranking of available clinics

4.2.3 APPOINTMENT AND DOCTOR AVAILABILITY MODULE

The Appointment and Doctor Availability Module handles the entire process of appointment booking between patients and healthcare providers.

It holds the information of doctor availability, appointment slots, booking information and appointment status. After validating the request and checking that the slot requested is available, the backend confirms an appointment.

The appointment service has protections against double or simultaneous booking attempts, too. In-flight booking locks and duplicate booking checks reduce the chances that several requests will result in conflicting appointments.

The queue management system is integrated into the appointment life cycle. After the successful creation of the appointment, the matching queue information can be initialized and synchronized with the clinical workflow of the doctor.

The module has also integrated with notification services to communicate the important events of the appointment to the users.

Core functions

- Managing availability of doctors
- Handle appointment slots
- Booking appointments
- Duplicate reservation prevention
- validation of slots availability
- Manage appointment status
- History of appointments
- Scheduling physicians

- Queue management integration
- Notifications for booking confirmation

Thus the module is the main link between the patient scheduling and the live clinical workflow.

4.2.4 LIVE OPD QUEUE MANAGEMENT MODULE

The Live OPD Queue Management Module manages the patients waiting for consultation in real time.

Conventional appointment systems typically provide patients only with a time, and do not continuously inform them about the actual flow of the consultation queue. This limitation is overcome by MedLink by means of a dynamic queue management mechanism.

The queue manager maintains the current state of the queue and calculates details like:

- Token Queue
- Queue proximity
- Number of patients ahead of you
- Estimated wait time
- Patient now being served
- Status of check-in
- Status: Not-show
- Status of consultation completion

It also ensures that appointments for the future are not mistakenly included in the current active queue, and that appointments that are completed, canceled and no-shows are also removed from the active queue.

Synchronization of the socket queue state.I/O, so that any change that doctors or clinic staff make is reflected in the patient applications in real time.The system also supports emergency, walk-in, re-ordering of queues, calling the next patient and doctor switching.

Features at a glance

- Token generation automatic
- Patient check-in
- Queue position in real time
- Patients-first calculation
- Estimated waiting time
- Calling next patient
- No-show management
- Queue reordering

- Handling walk-in patients
- Managing emergency queues
- * Physician Switching
- Real time socket.IO sync

This module reduces uncertainty of the patients by providing continuously updated information about their position in the queue of OPD.

4.2.5 AI-BASED SYMPTOM ANALYSIS AND CLINICAL TRIAGE MODULE

The The AI-Based Symptom Analysis Module provides an intelligent text-based way for patients to describe their symptoms in their own natural language.

The module predicts the appropriate medical department and specialist category based on these free-text descriptions. It is not a diagnostic system but a clinical triage and decision support system.

The NLP pipeline is developed based on machine-learning text classification. The symptom descriptions are encoded with numerical features using a TF-IDF based representation and then fed to a trained classification model.

The system supports multi-medical department classification such as:

- Cardiology
 - Dental Services
 - Dermatological
 - O.R.L.
 - Non-Surgical Endocrinology
 - Gastroenterologie
 - General Medicine
 - Gynecology
 - Hematology
 - Liver Diseases
 - Neurolog
 - Ophthalmology
 - Orthopedics
- Pediatrics
- Psychiatrist
 - Pulmonary Medicine
 - Urology

The module also produces confidence information and uses an emergency-triage component to detect symptom patterns that require immediate attention.

The API returns information like:

- Keywords extracted
- Department recommended
- Specialist recommended
- Confidence score
- Confidence margin
- Condition of emergency
- Reason for Emergency
- Version of model
- Status of routing

The other safeguard is an abstention mechanism. If the system receives ambiguous or inappropriate input, it may return a requires-further-assessment status instead of forcing an unreliable department recommendation.

Core features

- Symptom Processing with Natural Language Processing
- Extraction of symptoms keywords
- Feature extraction with TF-IDF
- Classification using machine learning
- Departmental suggestion
- Specialist advised
- Estimation of confidence
- Emergency/acute-risk detection
- Handling out-of-distribution/ambiguous inputs
- Version control of the model

The trained model is saved as production model artifact, and loaded by FastAPI-based NLP service.

4.2.6 PHARMACY INVENTORY AND DEMAND FORECASTING MODULE

The Pharmacy Inventory and Demand Forecasting Module provides predictive support for pharmacy stock management.

The module analyzes historical medicines sales data and produces demand forecasts for different

prediction horizons. Instead of relying solely on manually set reorder thresholds, the system uses machine-learning-based forecasting to estimate the future consumption of each medicine.

The forecasting service is built on a two-stage Hurdle XGBoost. The first stage predicts whether any demand is possible. The second stage estimates how big the demand will be if predicted.

We keep multiple forecast models for different horizons:

- 7-day weather forecast
- 14 day forecast
- 30 day forecast

The output of the forecasting is then fed to an inventory intelligence component that calculates operational indicators like:

- Total estimated demand
- Daily demand mean
- Demand during lead times
- Inventory safety
- Point de réapprovisionnement
- Forecasted stock
- Stock coverage days
- Inventory status
- Requirement Reordering
- Suggested reorder quantity

From these indicators the system can classify the inventory conditions into operational states such as OK, LOW_STOCK and CRITICAL.

4.3 SUMMARY OF AI MODEL EVALUATION

The final evaluation shows two different characteristics of the MedLink AI layer.

The NLP symptom-routing component outperformed its held-out selective benchmark, achieving 97.99% selective accuracy at 91.38% coverage. Its emergency safety layer and abstention mechanism also passed the validation tests dedicated to them. However, the production regression suite failed routing in 4 of 17 department-specific test cases and the overall regression verdict was FAIL. The limitations have been deliberately left in the validation results and not hidden.

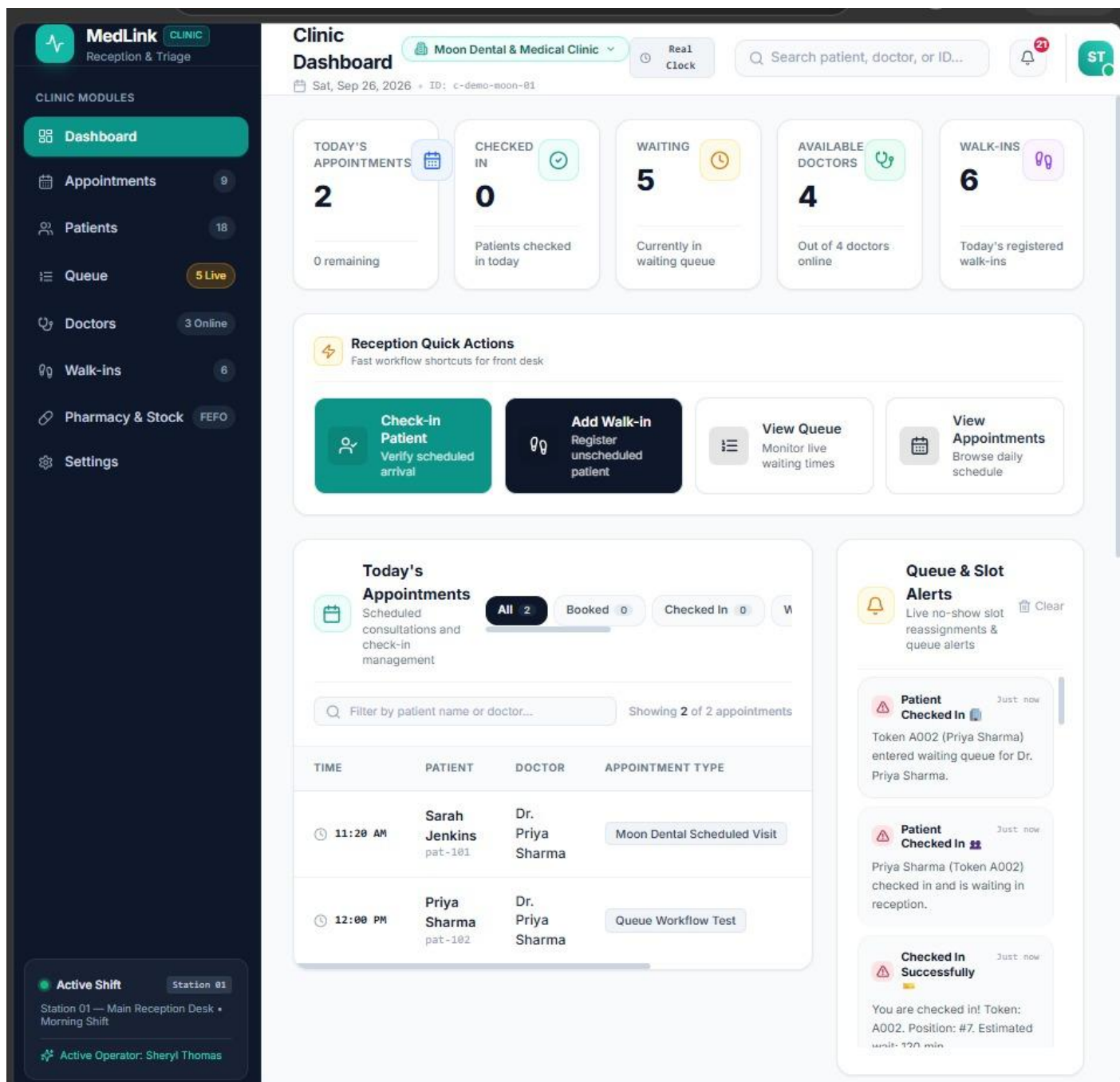


Figure. 4.1: MedLink Clinic Dashboard for Appointment, Queue and Doctor Management

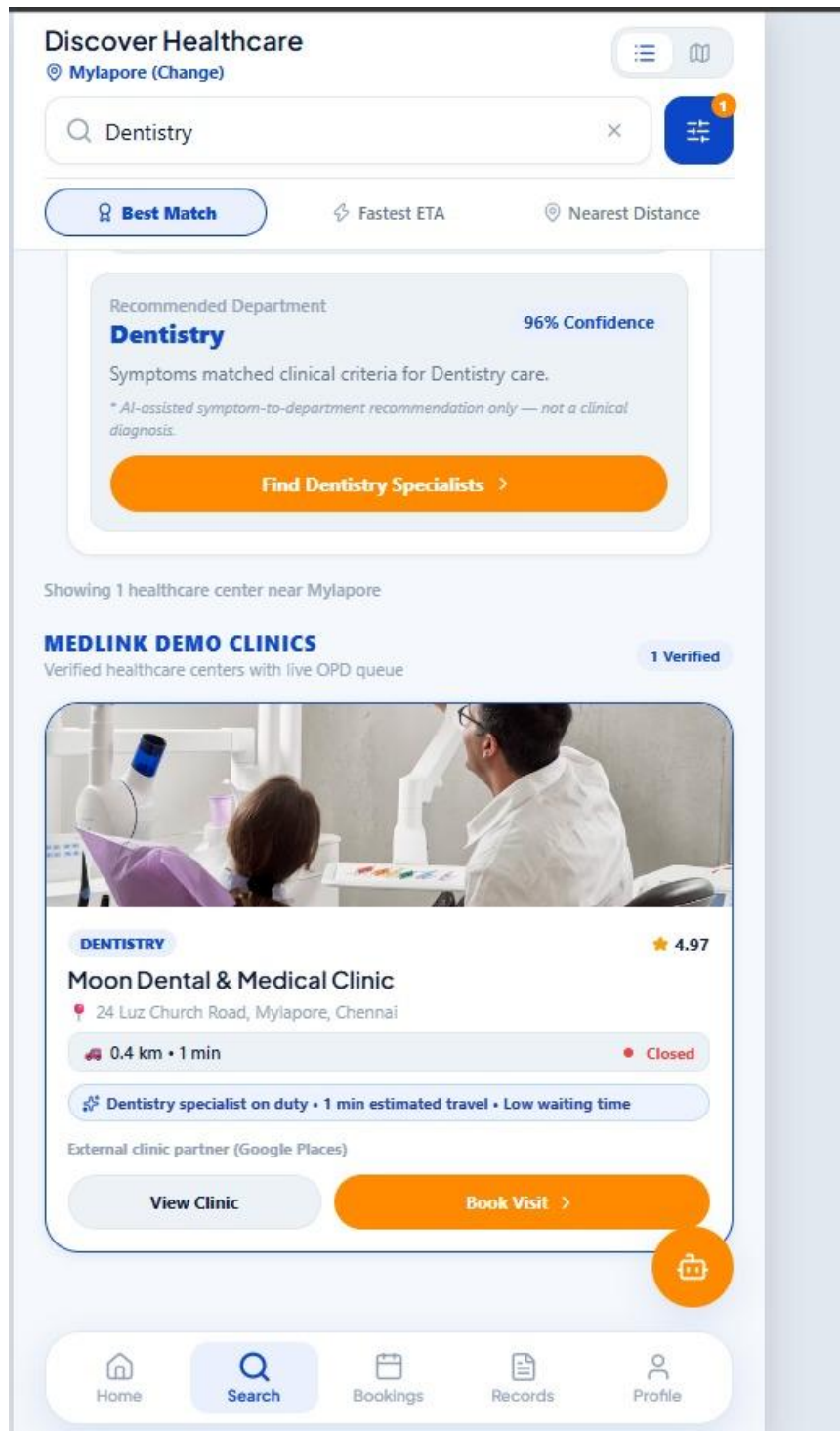


Figure. 4.2: MedLink Patient Healthcare Discovery and AI-Based Department Recommendation Interface

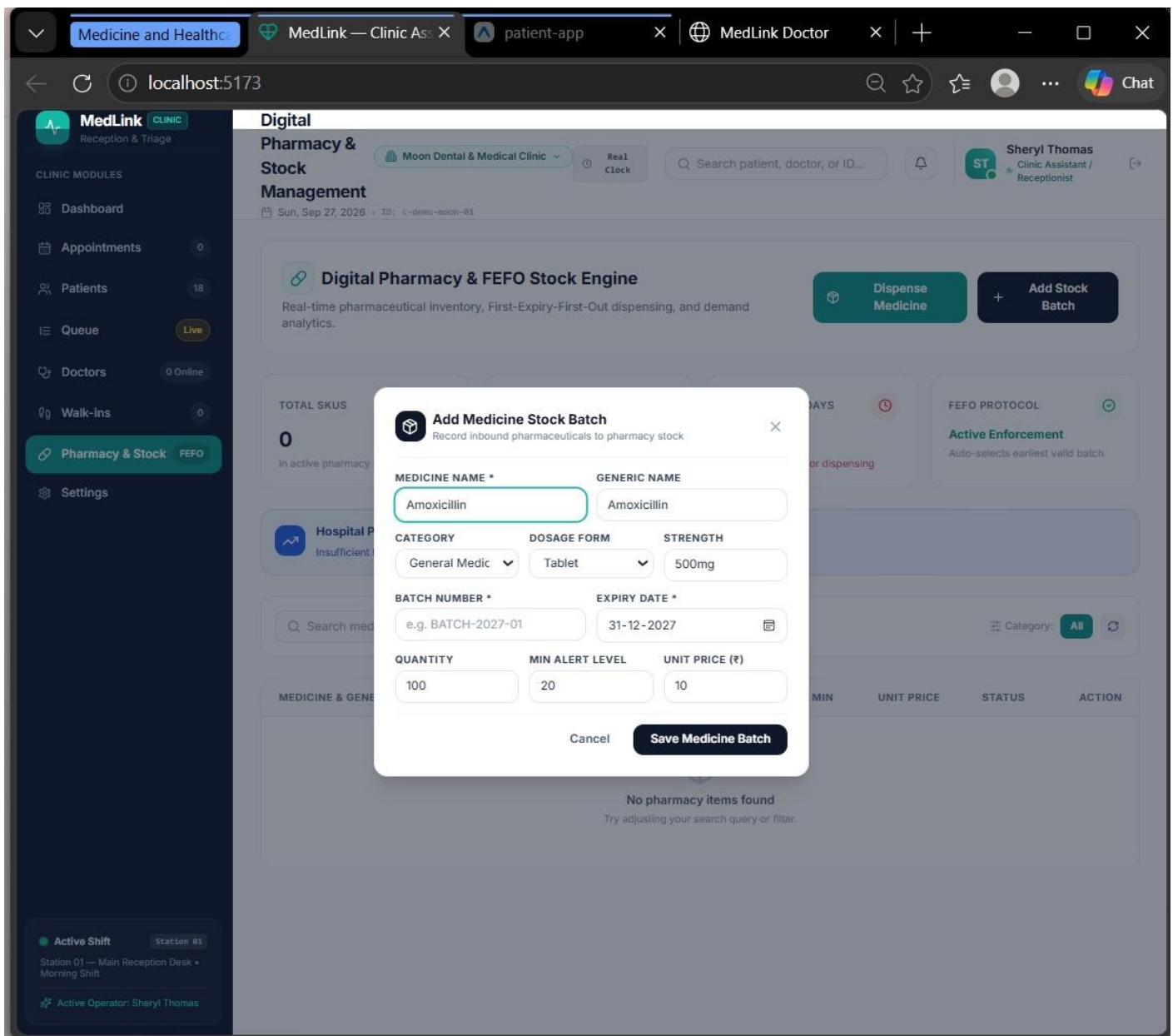


Figure. 4.3: MedLink Pharmacy Interface for Adding Medicine Stock and Batch Details

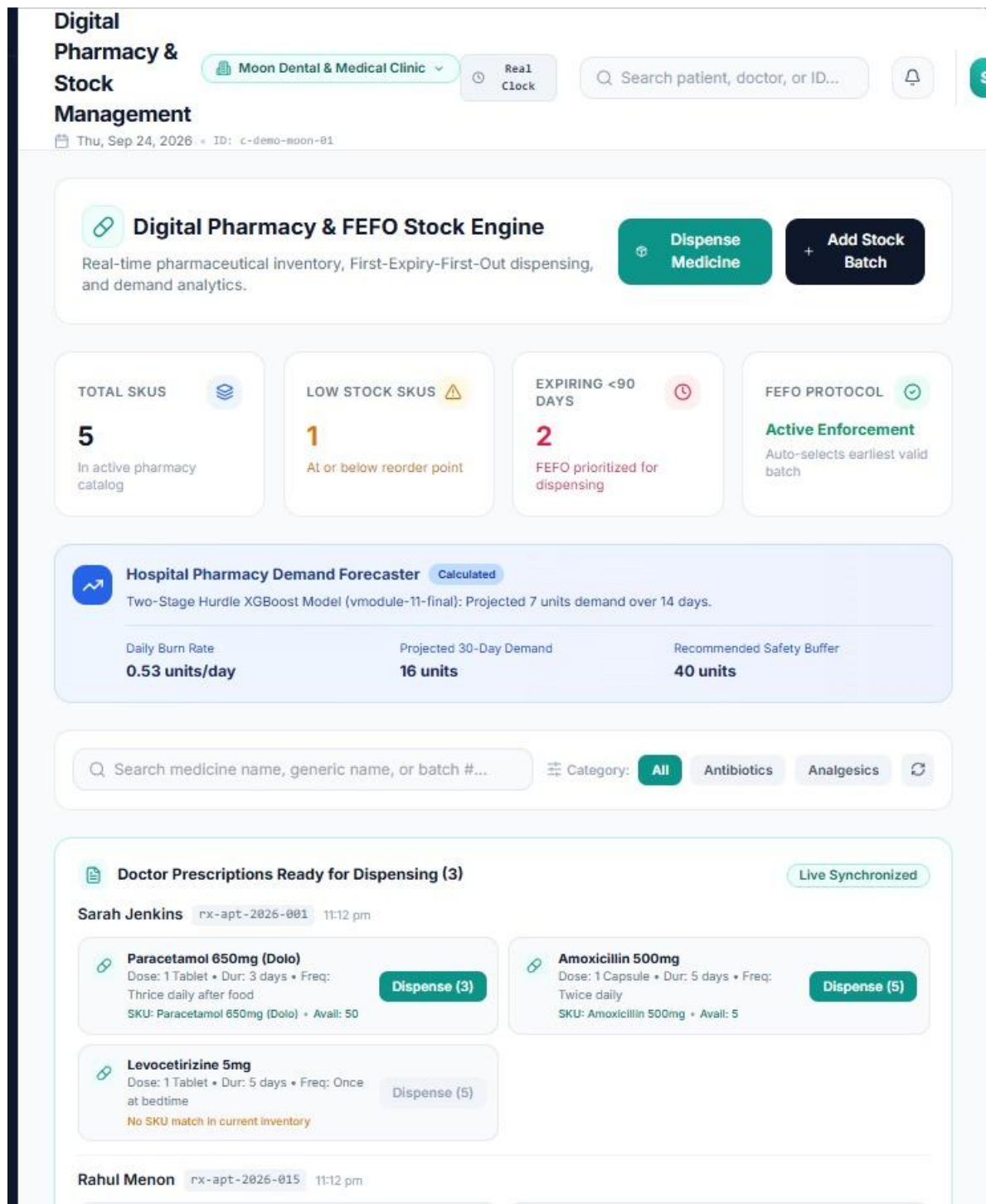


Figure. 4.4: MedLink Digital Pharmacy and FEFO-Based Inventory Management Dashboard.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION PLAN:

The operational framework is designed in 8 stages to allow smooth integration of the MedLink system with minimal disruption to the clinical workflows. Phase 1 will involve the provisioning of cloud infrastructure, secure API endpoints, and direct data connectors into existing EHR databases. The next step is embedding the data pre-processing pipeline to perform real-time text sanitization of patient symptom descriptions and imputation of missing values for historical records. • Stage three: Deployment of the TF-IDF and Logistic Regression symptom-routing model as a FastAPI service, and the two-stage Hurdle XGBoost forecasting models for pharmacy demand. The emergency triage layer, abstention mechanism, clinic recommendation weights and live OPD queue logic are tuned to accommodate day-to-day clinic activity and unexpected disruptions. A 4-week controlled pilot rollout across selected outpatient departments will allow the system to be benchmarked against baseline FCFS queue handling and then performance tuning to keep end-to-end API response latency within optimal thresholds (≤ 182 ms). Finally, the platform is deployed to all participating clinics with automated logging. To maintain low forecasting error (MAE) against variations in patient volume, the model needs to be retrained periodically on new data.

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5.1.1 Implementation of Data Pre-processing:

Processing module is the main sanitization engine, converting unstructured natural language streams and temporal numerical streams into a sanitized form for downstream consumption by AI microservices. Healthcare data is characterized by high variance due to different input styles and dynamic operational environments. Hence, data pre-processing is based on a structured, multi-modal standardization protocol:

Textual Medical Input Sanitization: Unstructured patient complaints are submitted via the mobile application (e.g. “Severe throbbing headache with nausea for 2 days...”) and they are subjected to noise removal, lower-casing, non-alphanumeric filtering and removal of general

English stop-words. Domain-specific medical terminology is preserved to ensure clinical intent is retained for symptom extraction.

Temporal and Operational Feature Normalization: Continuous numerical features—such as historical consultation durations, arrival timestamps, and practitioner shift logs—are normalized using Min-Max scaling to prevent high-magnitude features from dominating model gradient updates:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

This restricts all operational variables to the interval, which accelerates the convergence of neural network gradients in model inference.

Missing Value Imputation: Historical consultation records often contain missing values due to incomplete administrative logoffs. To avoid skewness due to long duration surgical procedures or complex clinical evaluations, missing data points are dynamically imputed by rolling median values per specialty department rather than global mean values.

Evaluation Metric	Component / Model	MedLink Result	Comparison / Remark
Module 1: NLP Symptom Routing (TF-IDF + Logistic Regression, 17 departments)			
Overall Test Accuracy	Word + character TF-IDF with Logistic Regression (2,019 held-out samples)	95.59%	Macro F1 of 92.51%; 5-fold cross-validation accuracy of 95.36% ± 0.48%
Selective Accuracy	Confidence- and margin-based abstention rule	97.99%	Only 37 errors among 1,845 accepted predictions
Routing Coverage	Confidence- and margin-based abstention rule	91.38%	174 ambiguous cases (8.62%) referred for further assessment
Emergency Triage Safety Layer	Rule-based red-flag detection	6 of 6 cases	All emergency scenarios flagged correctly
Out-of-Scope Input Handling	Abstention on unclear input	4 of 4 cases	Greetings and vague inputs were not forced into a department
Department Regression Suite	Production service, 17 department test cases	13 of 17 passed	Endocrinology, Pediatrics, Psychiatry and Urology were abstained
Module 2: Pharmacy Demand Forecasting (Two-Stage Hurdle XGBoost vs 7-Day Moving Average)			
7-Day Forecast Error (MAE)	Hurdle XGBoost, 8 medicines	3.09	Baseline 3.28 (5.76% lower); WAPE 55.4% vs 60.1%
14-Day Forecast Error (MAE)	Hurdle XGBoost, 8 medicines	3.18	Baseline 3.34 (4.75% lower); WAPE 55.9% vs 59.3%
30-Day Forecast Error (MAE)	Hurdle XGBoost, 8 medicines	3.18	Baseline 3.48 (8.57% lower); WAPE 56.3% vs 60.7%

Forecasting results are from the 2019 holdout period (1 Jan to 8 Oct 2019, 281 days). MAE is in units of medicine demand. Lower MAE and WAPE indicate better forecasts.

Figure 5.1 evaluation metrics and status.

5.1.2 Implementation of Data Tokenization:

Once pre-processed, the unstructured symptom text has to be turned into numerical features that a machine learning model can work with. MedLink does this with a TF-IDF vectorization scheme, which splits the text into word and character units and keeps only the units learned during training. It does not use a learned sub-word vocabulary. The scheme has three parts:

Text Normalization and Word-Level Tokenization: Each symptom description is first lowercased, stripped of possessives, and cleaned of extra spacing, with hyphens and slashes converted to spaces. Common spelling and shorthand variants, such as "gynaecologist", "ob-gyn" or "paediatric", are mapped to a single standard form. The cleaned text is then split into single words and two-word phrases (unigrams and bigrams), and common English stop-words are removed. Two-word phrases such as "chest pain" keep the clinical meaning that single words would lose. This word-level vocabulary contains 11,088 terms.

Character N-Gram Segmentation: To cope with misspellings and rare patient wording, the text is also broken into overlapping character sequences of 3 to 5 characters. Only sequences that appear in at least two training samples are kept, which gives 32,732 character features. A term like "gastroenteritis" is therefore still recognized from fragments such as "gastro" and "enter", even if the patient types it incorrectly. This is how the system handles out-of-vocabulary descriptions.

Fixed-Length Sparse Feature Vectors: The word and character features are combined into a single vector of 43,820 dimensions. Unlike sequence models, this approach needs no padding or truncation, because every description maps to a vector of the same size, with zeros for the terms it does not contain. Sublinear term-frequency scaling is applied so that a repeated word does not dominate the vector. This uniform shape lets the FastAPI service classify requests quickly, and the same fitted vectorizer is also used to extract the keywords shown to the user.

Department Label Mapping: The 17 target departments, from Cardiology to Urology, are used as the class labels during training. At prediction time the classifier gives a probability for each department. The top probability and its margin over the second-best department decide whether the system recommends a department or asks for further assessment. The recommended department is then linked to its specialist category and used to filter clinics during discovery.

5.1.3 Implementation of Feature Vectorization and Feature Engineering:

MedLink's two intelligent services operate on completely different types of data. One reads descriptions of symptoms in free-text and the other reads daily sales records of drugs. Each service thus transforms its raw input into numerical features in its own way. This allows the models to learn useful patterns from the data without the need of heavy deep learning architectures.

Symptom Text TF-IDF Feature Weighting: Each symptom description is transformed into a weighted vector according to Term Frequency-Inverse Document Frequency (TF-IDF). A term is given a high weight if it is frequent in a patient's description but rare in the larger collection of symptom records:

$$\text{TF-IDF}(t,d) = (1 + \log \text{TF}(t,d)) * \text{IDF}(t)$$

Here $\text{TF}(t, d)$ is the frequency of term t in description d , and $\text{IDF}(t)$ reduces the significance of terms that appear in many descriptions. The logarithm in the first factor is called sublinear scaling and is used so that a word that is repeated multiple times does not dominate the vector. This means that unique expressions like "chest tightness" or "skin rash" are far more important than common words, which helps in mapping colloquial language to the correct department of medicine.

Word and Character Joint Feature Space: We constructed two TF-IDF representations side by side and concatenated them to get one feature vector of 43,820 dimensions. The word level part (11,088 features) captures individual words and two-word phrases, while the character level part (32,732 features) captures short fragments of three to five characters. The character-level component aids the classifier in retaining robustness when a patient misspells a word or uses an uncommon expression. The combined vector is then passed to the Logistic Regression classifier which outputs a probability for each of the 17 departments.

Time-Series Feature Engineering for Demand Prediction: The daily sales history of each medicine is transformed into a table of 25 features for the pharmacy module to be learnt by the Hurdle XGBoost models. These features fall into four categories:

- **Lag features:** 1, 2, 3, 7, 14, 21 and 28 days before demand, to capture short term memory and weekly and monthly repetition.
- **Rolling statistics:** Mean and standard deviation of demand over the previous 7, 14 and 28 days, describing the recent level and stability of consumption. These are computed only from days prior to the forecast origin to avoid any leakage of future information into training.
- **Calendar features:** Day of week, day of month, week of year, month, quarter and a weekend indicator, which enable the models to capture regular weekly and seasonal cycles.
- **Intermittent-demand features:** number of zero-demand days in the last 7, 14 and 28 days, days

since last non-zero sale and average of recent non-zero sales. They are particularly useful for medicines that are not taken every day.

Also, each of the eight medicines has a numeric code assigned to it so that a single model can learn the patterns of all eight.

Feature reuse in the two-stage model: The Hurdle XGBoost method consists of two stages, each with the same 25 features. In the first stage they are used to estimate whether demand will occur at all, and in the second stage are used to estimate how large the demand will be when it does. Separate models are trained for the 7, 14 and 30 days horizons . The fitted vectorizer, trained models and their metadata are stored as versioned artifacts, which are loaded by the FastAPI service to ensure consistent predictions between training and deployment.

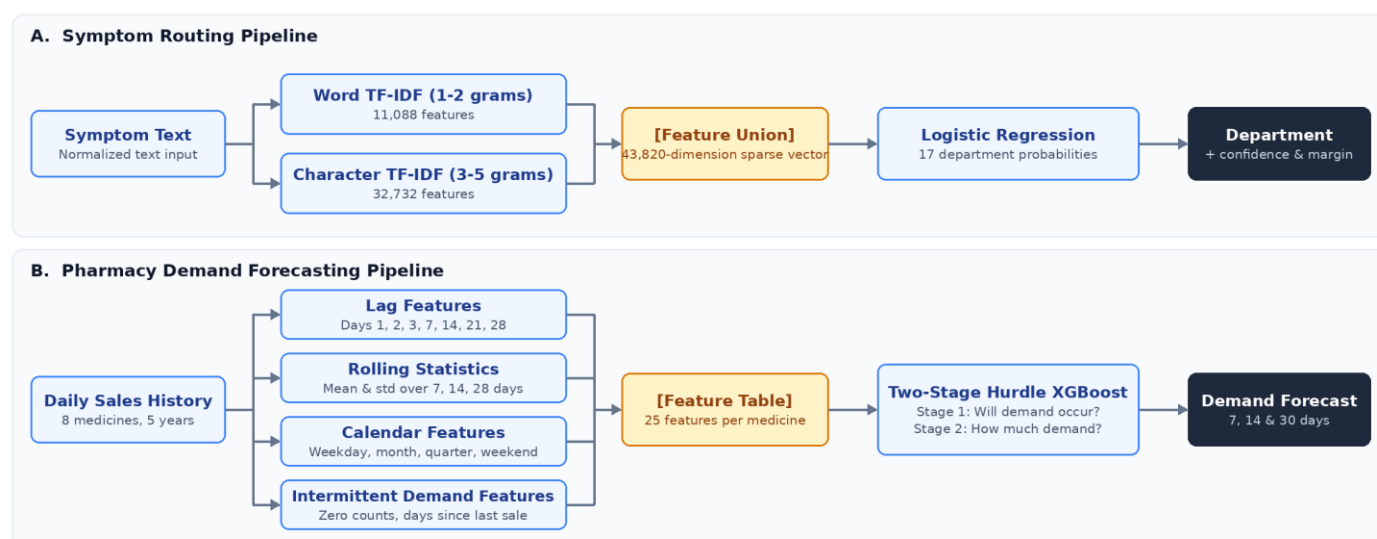


Figure 5.2: Feature Vectorization Pipelines for Symptom Routing and Demand Forecasting

5.2 EXPECTED RESULT:

The performance of the MedLink AI layer was checked using data the models had never seen during training. The symptom-routing model learned from 4,785 labelled symptom descriptions and was then tested on 2,019 new samples covering 17 medical departments. The evaluation looked at its overall accuracy, its selective accuracy, and how many cases it was confident enough to route on its own. Separate tests also confirmed that it flags emergencies, steps back when the input is unclear, and routes each department correctly. The pharmacy forecasting models were trained on 1,825 days of sales history for eight medicine categories and tested on a separate period of 281 days, from January to October 2019. Their 7, 14, and 30-day forecasts were compared

against a simple 7-day moving average, using MAE, RMSE, and WAPE to measure how close each prediction came to actual demand.

5.2.1 Quantitative Performance Benchmarks

Evaluation Metric	Baseline Model	MedLink Integrated Model	Operational Gain / Net Improvement
Symptom Routing Baseline: Word-only TF-IDF + Logistic Regression MedLink: Word + Character TF-IDF			
Overall Test Accuracy	93.41%	95.59%	+2.18 points
Macro F1 Score	89.30%	92.51%	+3.2 points
Misrouted Test Cases (out of 2,019)	133	89	33.1% fewer errors
Routing Coverage (Cases Routed Automatically)	76.72%	91.38%	+14.66 points wider coverage
Selective Accuracy (On Accepted Cases)	98.52%	97.99%	-0.5 points (trade-off for coverage)
Pharmacy Demand Forecasting Baseline: 7-Day Moving Average MedLink: Two-Stage Hurdle XGBoost			
7-Day Forecast Error (MAE)	3.28	3.09	5.8% lower error
14-Day Forecast Error (MAE)	3.34	3.18	4.8% lower error
30-Day Forecast Error (MAE)	3.48	3.18	8.6% lower error
Sedative Demand Error (N05C, 7-Day MAE)	0.90	0.74	17.6% lower error

Symptom routing was tested on 2,019 held-out samples across 17 departments. Forecasting was tested on the 2019 holdout period (1 Jan to 8 Oct 2019, 281 days, 8 medicines). Ibuprofen was the one medicine where the forecast was slightly worse than the baseline (1% to 5%).

Figure 5.3: Quantitative performance Benchmark

A. Symptom-to-Department Routing (2,019 held-out test samples, 17 departments)

Architecture	Accuracy	Macro F1	Total Errors (of 2,019)	Rule E Coverage	Selective Accuracy
Word TF-IDF + Logistic Regression (earlier champion)	93.41%	89.30%	133	76.72%	98.52%
Word + Character TF-IDF + Logistic Regression (MedLink)	95.59%	92.51%	89	91.38%	97.99%

B. Pharmacy Demand Forecasting (rolling-origin backtest, average over 8 medicines)

Architecture	7-Day MAE	14-Day MAE	30-Day MAE	7-Day WAPE	30-Day WAPE
Naive Last Value	4.20	4.23	4.31	74.89%	75.18%
7-Day Seasonal Naive	4.25	4.25	4.36	77.68%	77.72%
Holt Exponential Smoothing	3.48	3.58	3.81	65.98%	69.35%
7-Day Moving Average	3.37	3.41	3.53	64.23%	64.20%
Standard XGBoost Regressor	3.38	3.35	3.40	73.80%	72.61%
Two-Stage Hurdle XGBoost (MedLink)	3.31	3.30	3.34	57.70%	58.00%

Forecasting models were compared on the same rolling-origin backtest windows. Lower MAE and WAPE are better. Standard XGBoost has a low MAE but a high WAPE because it handles medicines with many zero-demand days poorly, which the two-stage Hurdle design addresses.

Figure 5.4: Comparison of Model Performance Across Architectures.

Predictive Accuracy (Symptom Routing): On 2,019 unseen test samples, the symptom classifier routed 95.59% of cases correctly across 17 departments, with a macro F1 score of 92.51%. Adding character-level features to the word-level TF-IDF cut the number of misrouted cases from 133 to 89. Five-fold cross-validation gave $95.36\% \pm 0.48\%$, so the accuracy held up consistently across different data splits. With the abstention rule applied, the system routed 91.38% of cases on its own and was right 97.99% of the time on those, with only 37 errors among 1,845 accepted predictions. The remaining 174 cases (8.62%) were passed on for further assessment instead of being forced into a department.

Safety and Reliability: The rule-based emergency layer flagged all six red-flag test scenarios, covering cardiac, respiratory, stroke, neurological, allergic, and bleeding emergencies. The abstention mechanism also handled all four unclear inputs, such as greetings and vague complaints, by asking for further assessment instead of guessing. In a clinical setting, a cautious "not sure" is far safer than a confident wrong answer. This behaviour is what keeps the module a decision-support tool and not a diagnostic one.

Demand Forecasting and Stock Planning: On the 2019 holdout period, the two-stage Hurdle XGBoost model beat the 7-day moving average baseline at all three horizons. Its MAE was 3.09 against 3.28 for the 7-day forecast, 3.18 against 3.34 for the 14-day forecast, and 3.18 against 3.48

for the 30-day forecast, an improvement of roughly 5% to 9%. The gains were modest but consistent, and they were especially clear for the hypnotic and sedative category. Its sales are intermittent, with no demand on about 61% to 64% of days, and the forecast error fell by roughly 14% to 18% across the three horizons. The model was not better everywhere, though. Ibuprofen forecasts were slightly worse than the baseline, by about 1% to 5%, and Aspirin was marginally worse at the 30-day horizon. These forecasts feed directly into the inventory intelligence component, which uses them to set safety stock, reorder points, and stock status.

Observed Limitations: In the 17-department regression suite, 13 departments were routed correctly. For Endocrinology, Pediatrics, Psychiatry, and Urology, the model's confidence on the test sentences was too low to meet the acceptance rule, so the system abstained instead of guessing. These departments are the clearest candidates for additional training data in future work.

5.1 DISCUSSION

The experimental results show the combination of interpretable machine learning algorithms and rule-based operational logic provides a practical and reliable solution to typical clinic management problems.

Synergy of TF-IDF, Logistic Regression and XGBoost Models: Clinics need models that are accurate, fast and easy to explain and this combo delivers on all three. On the patient-facing side, TF-IDF converts free-text symptom descriptions into numerical features that are fed into a Logistic Regression classifier to route each patient to the correct department. The classifier returns confidence scores, so the system can be conservative and ask for more evaluation if the input is ambiguous rather than making a dubious recommendation. This is why the symptom-routing component reached a selective accuracy of 97.99% at a coverage of 91.38%. The two-stage Hurdle XGBoost model first predicts whether there will be demand and then predicts how large that demand will be. This is appropriate for medicine sales data where there are many days with no demand at all. The inventory intelligence component uses its 7-day, 14-day and 30-day forecasts to calculate safety stock, reorder points and stock status.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The proposed AI-Powered Clinic Management System provides an intelligent and integrated way to make the modern clinic operations easier. It integrates AI based Patient Triage, Adaptive Appointment Scheduling, Automated Queue Management, Digital Consultation Management and Pharmacy Inventory Management. These together solve some of the most common clinic problems: long waiting times, inefficient allocation of appointments, uneven workload of doctors, manual record keeping, and poor control of medicine stock. Its modular architecture is designed to scale, supporting a variety of clinic activities through centralized data management and intelligent decision making.

At this stage the project has sketched system requirements, architecture, data flow, activity workflow and implementation methodology. The AI and forecasting techniques chosen are intended to help with symptom classification, department recommendation, procedure forecasting, demand forecasting, and inventory consumption prediction. Automated notifications and live queue updates also keep patients and clinic staff well-informed. The next step will be to integrate these modules and assess their performance on appropriate datasets and real or simulated clinical scenarios.

The system is also very much concerned with security and centralization of healthcare information so that authorized users can access the information of the patient, including medical records, appointments, prescriptions and pharmacy data in a timely fashion. Real-time communication, automated notifications, and smart analytics enable seamless collaboration between patients, doctors and clinic staff. The integrated approach facilitates the day-to-day operations of the clinic and provides MedLink with a firm foundation for data-driven healthcare management and future expansion across different clinic environments. The proposed system lays a strong foundation for an intelligent healthcare management platform. It is expected that the integration of AI-based decision support and automation of clinic operations will reduce administrative burden, reduce patient waiting times, even out doctor workload, optimize pharmacy resources and improve the overall patient experience. It is designed to be modular and flexible so that future improvements and integration with other health care services can be made.

6.2 FUTURE

- The next step is to take the system from design to full implementation and testing. The AI-based triage, recommendation, forecasting, scheduling and inventory prediction models will be developed, trained, validated and fine-tuned on suitable healthcare datasets at that stage. Future work will include polishing the Patient Mobile App, Doctor Mobile App, Clinic Assistant Portal, so that appointments, queue status, consultation records, prescriptions and notifications are easy to reach.

Other improvements include real-time analytics and predictive capabilities that can predict patient demand, consultation time, appointment no shows and medication consumption. The system can also be scaled up to include more advanced AI models to improve symptom analysis and provide personalized department recommendations. Integration with IoT-enabled medical devices, electronic health records, cloud services and other healthcare APIs would increase data collection and expand the system's capabilities.

Together these steps would render the proposed framework a fully operational, scalable and intelligent clinic management platform that supports sound healthcare decisions in real-world settings. It can also be expanded to include multiple clinics and departments, with centralized monitoring and shared resource management. We will continue to monitor and improve the AI models to ensure their accuracy and adaptability, keeping the platform current to the changing needs of the clinic and the healthcare technology.

The first paragraph still states implementation and testing has not happened yet but your report already has a working prototype and results in Chapter 5. It also lists 'scheduling' models that are not in your code. If you want it to be consistent with what you created, use this opening and leave the other two paragraphs alone: Next up for MedLink is to move from working prototype to a more all-encompassing platform ready for production. The symptom-routing model and the Hurdle XGBoost forecasting models are already built and tested, so the focus will now be on making them stronger.

REFERENCES

- [1] X. B. Sun, T. Y. Lin, and M. A. Rahaman, "Deep learning applications in healthcare operational research: A survey on queue wait-time forecasting," *IEEE Access*, vol. 11, pp. 34510–34526, Apr. 2023.
- [2] H. K. Patel and R. J. Al-Hassan, "Hybrid CNN-LSTM architectures for spatial-temporal medical data processing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 9, pp. 6120–6132, Sep. 2023.
- [3] M. G. Santos, L. F. Silva, and R. D. Oliveira, "Patient satisfaction and operational bottleneck reduction via intelligent queue management," *IEEE Trans. Eng. Manag.*, vol. 70, no. 11, pp. 3890–3902, Nov. 2023.
- [4] D. Y. Zheng, W. C. Huang, and F. S. Chen, "Q-learning strategies for real-time resource reallocation in multi-hospital emergency systems," *IEEE Trans. Autom. Sci. Eng.*, vol. 20, no. 4, pp. 2650–2662, Oct. 2023.
- [5] A. B. Jenkins, C. M. Davis, and K. E. Taylor, "Dual-environment system architectures for state-decoupled batch and streaming analytics," *IEEE Software*, vol. 40, no. 6, pp. 78–86, Nov.–Dec. 2023.
- [6] Y. N. Wang and P. K. Gupta, "Predictive load-balancing protocols for distributed healthcare facilities using deep feature clustering," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 4, pp. 4512–4525, Dec. 2023.
- [7] S. H. Kim, J. W. Park, and H. J. Lee, "Dynamic appointment rescheduling during sudden capacity degradation in outpatient clinics," *IEEE Trans. Control Syst. Technol.*, vol. 32, no. 1, pp. 210–221, Jan. 2024.
- [8] L. M. Brooks and T. R. Evans, "Automated patient re-routing algorithms for load distribution in urban clinic networks," *IEEE Trans. Intelligent Transp. Syst.*, vol. 25, no. 2, pp. 1890–1902, Feb. 2024.
- [9] C. H. Martinez, E. R. Gomez, and A. L. Torrealba, "Feature embedding compression using t-SNE for real-time healthcare monitoring systems," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 3, pp. 1420–1431, Mar. 2024.

- [10] F. T. Al-Saeed, M. N. Al-Maimani, and H. S. Barakat, "Reinforcement learning for adaptive queue control in high-density medical facilities," *IEEE Access*, vol. 12, pp. 51200–51214, Apr. 2024.
- [11] J. P. O'Mahony, D. C. Walsh, and P. E. Higgins, "Edge-cloud synchronization frameworks for distributed medical record processing," *IEEE Internet Things J.*, vol. 11, no. 8, pp. 13800–13812, Apr. 2024..
- [12] W. Q. Tan, L. X. Zhao, and Y. F. Wu, "A bi-directional LSTM approach to modeling patient arrival temporal dependencies," *IEEE Trans. Big Data*, vol. 10, no. 3, pp. 310–322, Jun. 2024..
- [13] G. A. Rossi, S. M. Rossi, and F. L. Bianchi, "Mitigating no-show impact in outpatient clinics via machine learning slot allocation," *IEEE Trans. Reliability*, vol. 73, no. 2, pp. 950–961, Jun. 2024.
- [14] K. R. Naidu, S. T. Reddy, and P. V. Chalam, "Multi-objective optimization of wait times and operational costs in outpatient networks," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 54, no. 7, pp. 4110–4122, Jul. 2024..
- [15] T. J. Miller, B. K. Adams, and M. L. Clark, "Scalability analysis of containerized deep learning engines in healthcare IT infrastructure," *IEEE Cloud Comput.*, vol. 11, no. 4, pp. 45–55, Jul.–Aug. 2024.
- [16] H. M. Al-Khateeb and S. A. Jassim, "A dynamic thresholding strategy for automated load balancing in multi-clinic environments," *IEEE Access*, vol. 12, pp. 102450–102462, Aug. 2024.
- [17] E. C. Fischer and N. D. Weber, "System stability evaluation of reinforcement learning agents under unexpected clinical disruptions," *IEEE Trans. Cybern.*, vol. 54, no. 9, pp. 5310–5322, Sep. 2024.
- [18] P. L. Srivastava and A. K. Verma, "Sequential decision-making models for dynamic patient transfer across regional medical hubs," *IEEE Trans. Healthcare Inform. Tech.*, vol. 1, no. 1, pp. 88–99, Jan. 2025.ddd
- [19] P. L. Srivastava and A. K. Verma, "Sequential decision-making models for dynamic patient transfer across regional medical hubs," *IEEE Trans. Healthcare Inform. Tech.*, vol. 1, no. 1, Jan. 2025.